

Program: Diploma in Civil and Rural Engineering	
Course Code: 6361A	Course Title: Water supply and Sanitation Engineering
Semester: 6	Credits: 4
Course Category: Program Core	
Periods per week: 4 (L:4, T:0, P:0)	Periods per semester: 60

Course Objectives:

- To impart knowledge to the students about the estimation of water demands for proper design of water supply works.
- To make the students aware about the different water conveyance and distribution systems.
- To Familiarize the students with different stages of water treatment, different methods of sewage treatment and disposal and operations involved.

Course Prerequisites:

Topic	Course code	Course name	Semester
Flow through Open Channel- Geometrical properties of channel section		Hydraulics and Irrigation	4
Basic knowledge in Chemistry		Engineering Chemistry	1

Course Outcomes:

On completion of the course, the student will be able to:

CO _n	Description	Duration (Hours)	Cognitive level
CO1	To identify the different sources of water and to determine water demands from available data for designing water supply systems.	14	Applying
CO2	Suggest appropriate water treatment methods	15	Understanding
CO3	Identify the components required for Building sanitation and Sewerage systems	14	Understanding
CO4	Suggest appropriate sewage treatment and disposal methods	15	Understanding
	Series test	2	

CO - PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3			3	3		
CO2		3		3	3		
CO3	3				3		
CO4			3	3	3		

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

On completion of the course, the student will able to:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	To identify the different sources of water and to determine water demands from available data for designing water supply systems.		
M1.01	Describe various surface and sub surface sources of water.	3	Understanding
M1.02	Determine water demands for a water supply scheme	4	Applying
M1.03	Outline standard tests for analyzing water for drinking purposes	4	Understanding
M1.04	Explain the different rural water supply schemes by the Government	3	Understanding
Contents: Sources of Water: Surface source - lakes, streams, rivers and impounded reservoirs, Yield from surface source - Underground sources - springs, wells, infiltration wells and galleries - Yield from wells - test for yield. Intake Structures, Definition and types, Factors governing the location of an intake structure, Types of intakes. Demand of water: Factors affecting rate of demand, Variations of water demands, Forecasting of population, Methods of forecasting of population, (Simple problems on forecasting of population), Design period, Estimating of quantity of water supply required for city or town. Quality of water: Need for analysis of water, Characteristics of water - Physical, Chemical and Biological, Testing of water for Total solids, hardness, chlorides, dissolved Oxygen, pH, Fluoride, Nitrogen and its compounds, Bacteriological tests, E coli, B coli index, MPN, Sampling of water, Water quality standards as per IS 10500. Water supply schemes - Objectives, components, Rural water supply schemes by Government-NRDWP-Jal Jeevan Mission, Atal Bhujal Yojana, Jal Sakthi Abhiyan KRWSA-Jalanidhi, Jala Subhiksha.			
CO2	Suggest appropriate water treatment methods		
M2.01	Explain methods of aeration, plain sedimentation, and sedimentation with coagulation.	3	Understanding
M2.02	Outline the construction and operation of slow sand, rapid sand and pressure filters.	3	Understanding
M2.03	Explain methods of disinfection and advanced treatment methods for the water in a given situation	3	Understanding
M2.04	Identify the different components of a conveyance system	3	Understanding

M2.05	Explain with sketches the methods of distribution of water in a given situation	3	Applying
	Series Test - I	1	
Contents: Purification of water: Objectives of water treatment, Aeration – objects and methods of aeration- Plain sedimentation, Sedimentation with coagulation, principles of coagulation, types of coagulants, Jar Test, process of coagulation, types of sedimentation tanks, Clariflocculator - Filtration -Mechanism of filtration- classification of filters: slow sand filter, rapid sand filter, pressure filter. Construction and working of slow sand and rapid sand filter, operational problems in filtration – Disinfection: Objects, methods of disinfection, Chlorination- Application of chlorine, forms of chlorination, types of chlorination practices, residual chlorine and its importance – Flow diagram of water treatment plants. (No design for treatment units). Introduction to water softening, iron removal and Deflouridation techniques-Nalgonda method. Advanced Water treatments-Electrolysis, Reverse Osmosis. Conveyance: Types of Pipes used for conveyance of water, choice of pipe material, Types of joints and valves - location and function on a pipe lane. Conservation of water- Rainwater Harvesting. Distribution of water: Methods of distribution of water- Gravity, pumping, and combined system, Service reservoirs - functions and types, Layouts of distribution of Water-Dead end system, grid iron system, circular system, radial system; their suitability, advantages and disadvantages. Hand pumps-types, installation, operation, and maintenance of hand pumps.			
CO3	Identify the components required for Building sanitation and Sewerage systems		
M3.01	Define Sewage, Sewerage, Sewer, refuse, sullage, and garbage	3	Remembering
M3.02	Identify the different building sanitary fittings and systems of plumbing	4	Understanding
M3.03	Explain different sewerage systems	4	Understanding
M3.04	Describe the various types of sewer appurtenances	3	Understanding

Contents:

Building Sanitation: Necessity of sanitation, Necessity to treat domestic sewage, Definitions - Sewage, sullage, types of sewage. Definition of the terms related to Building Sanitation- Water pipe, Rain water pipe, Soil pipe, Sullage pipe, Vent pipe. Building Sanitary fittings- Water closet – Indian and European type, flushing cistern, wash basin, sinks, Urinals. Traps-types, qualities of good trap. Systems of plumbing - one pipe, two pipe, single stack, choice of system. Principles regarding design of building drainage, inspection and junction chambers, their necessity, location, size and shape.

Systems of Sewerage and Sewer Appurtenances: Types of Sewers, Systems of sewerage, self-cleansing velocity and non-scouring velocity, Laying, Testing and maintenance of sewers, Manholes and Drop Manhole-component parts, location, spacing, construction details, Sewer Inlets, Street Inlets.

CO4	Suggest appropriate sewage treatment and disposal methods		
M4.01	Identify characteristics of sewage	3	Understanding
M4.02	Outline the functions and working of screens, skimming tanks, and grit chambers.	3	Understanding
M4.03	Describe various sewage treatment and disposal methods	4	Understanding
M4.04	Explain various low cost rural sanitation systems	5	Understanding
	Series Test - II	1	

Contents:

Analysis of sewage: Characteristics of sewage, B.O.D., C.O.D. and its significance, Central Pollution Control Board Norms for discharge of treated sewage, Objects of sewage treatment and flow diagram of conventional sewage treatment plant.

Treatment of Sewage: Screening, Types of screens, Grit removal, Skimming, Sedimentation of sewage, Aerobic and anaerobic process, Sludge digestion, trickling filters, Activated sludge process, Disposal of sewage, Oxidation Pond, Oxidation ditch, aerated lagoons, anaerobic lagoon.

Rural sanitation: Low cost rural sanitation systems-Privies of a conservancy system of sanitation- Bore hole privy, Aqua privy, Sulabh Shauchalya, Two pit pour flush water seal, Septic tank and soak pit, Recycling and Reuse of domestic waste. Composting (Nadep or Vermiculture), Biogas plants. Low cost community and individual Garbage disposal systems, Ferro-cement storage tanks. Rural sanitation programmes by Government of India, Swachh Bharat Mission-Objectives and strategies. Suchitwa mission.

Text / Reference:

T/R	Book Title/Author
T1	S.C. Rangwala: Water Supply and Sanitary Engineering; Charotar Publishing House
R1	Garg, S.K., Environmental Engineering Vol. I and Vol. II, Khanna Publishers
R2	K.N.Duggal : Elements of Public Health Engineering ; S.Chand & Co.
R3	Sharma S.C, Environmental Engineering, Khanna Publishing House, New Delhi2
R4	Birdie, G. S. and Birdie, J.S.Water Supply and Sanitary Engineering, Dhanpat Rai
R5	Desai, Vasant, Rural Development in India: Past, Present and Future: a challenge in the crisis, Himalaya Publishing House, Delhi.
R6	Singh, Katar, Rural Development Principles, Policies and Management, Sage Publications India Pvt Ltd.

Online Resources:

Sl.No	Website Link
1	http://nptel.ac.in/courses
2	http://www.vlab.co.in/ba-nptel-labs-civil-engineering
3	Document Published by Ministry of Rural development, Govt. of India, Ministry of Rural development.